

ORDERS AND ORDER_DETAILS MANAGEMENT SYSTEM

1 Database Design

1.1 Order Table

Entity: Order

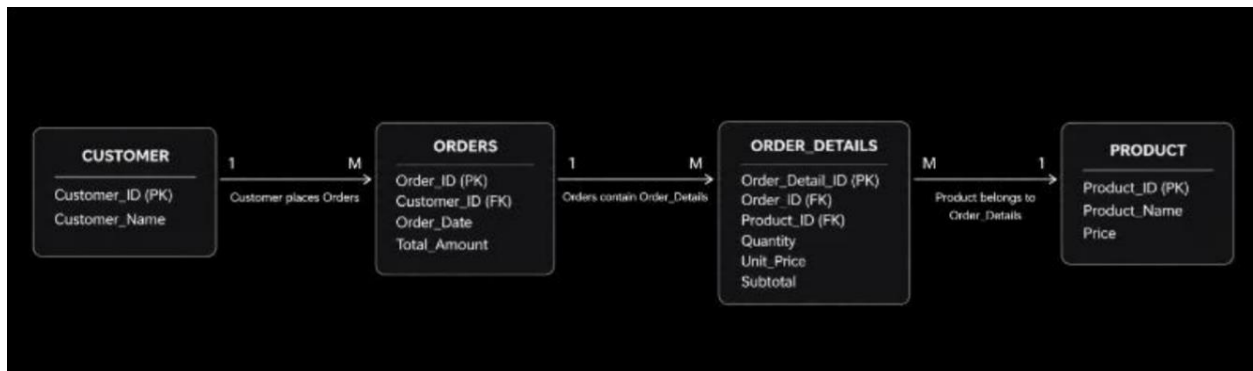
Attribute	Key type
Order_ID	Primary key
Customer_ID	Foreign key
Order_Date	-
Order_Status	-
Total_Amount	-

1.2 Order_Item Table

Entity:Order_Item

Attribute	Key type
OrderItem_ID	Primary key
Order_ID	Foreign key
Product_ID	Foreign key
Quantity	-
Unit_price	-

2.Table Relationship



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
  2     Order_ID NUMBER PRIMARY KEY,  
  3     Customer_ID NUMBER,  
  4     Order_Date DATE,  
  5     Total_Amount NUMBER(10,2),  
  6     Order_Status VARCHAR2(30),  
  7     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
  8 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Order_Details (  
  2     OrderDetail_ID NUMBER PRIMARY KEY,  
  3     Order_ID NUMBER,  
  4     Product_ID NUMBER,  
  5     Quantity NUMBER,  
  6     Unit_Price NUMBER(10,2),  
  7     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),  
  8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
  9 );
```

Table created.

4. Order Data Insertion

```

SQL> INSERT INTO Orders
  2 VALUES (401, 1, SYSDATE, 55000, 'Placed');

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (402, 2, SYSDATE, 25000, 'Placed');

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (403, 3, SYSDATE, 1598, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (404, 4, SYSDATE, 2500, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (405, 5, SYSDATE, 650, 'Placed');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

5. Order_Details Data Insertion

```

SQL> INSERT INTO Order_Details
  2 VALUES (501, 401, 101, 1, 55000);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (502, 402, 102, 1, 25000);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (503, 403, 103, 2, 799);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (504, 404, 104, 1, 2500);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (505, 405, 105, 1, 650);

1 row created.

```

6. Order Modification Operations

6.1 Order Status Modification

```
SQL> UPDATE Orders
  2 SET Order_Status = 'Shipped'
  3 WHERE Order_ID = 401;

1 row updated.
```

```
SQL> SELECT * FROM Orders
  2 ORDER BY Order_ID;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
401	1	24-SEP-26	55000	Shipped
402	2	24-SEP-26	25000	Placed
403	3	24-SEP-26	1598	Delivered
404	4	24-SEP-26	2500	Shipped
405	5	24-SEP-26	650	Placed

6.2 Order Amount Modification

```
SQL> UPDATE Orders
  2 SET Total_Amount = 1598
  3 WHERE Order_ID = 403;

1 row updated.
```

7. Order_Details Modification

```
SQL> UPDATE Order_Details
  2 SET Quantity = 3
  3 WHERE OrderDetail_ID = 503;

1 row updated.
```

8. Customer Order History Report

```
483          3 24-SEP-26      858 Placed

SQL> SELECT
2      c.Customer_ID,
3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
4      o.Order_ID,
5      o.Order_Date,
6      p.Product_Name,
7      od.Quantity,
8      od.Unit_Price,
9      o.Total_Amount,
10     o.Order_Status
11 FROM Customer c
12 JOIN Orders o
13     ON c.Customer_ID = o.Customer_ID
14 JOIN Order_Details od
15     ON o.Order_ID = od.Order_ID
16 JOIN Product p
17     ON od.Product_ID = p.Product_ID
18 ORDER BY c.Customer_ID, o.Order_ID;

CUSTOMER_ID
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT
-----
PRODUCT_NAME
-----
QUANTITY UNIT_PRICE TOTAL_AMOUNT ORDER_STATUS
-----
1
Renu Srinivasan
401 24-SEP-26

CUSTOMER_ID
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT
-----
PRODUCT_NAME
-----
QUANTITY UNIT_PRICE TOTAL_AMOUNT ORDER_STATUS
-----
Laptop
1      55000      55000 Shipped

CUSTOMER_ID
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT
-----
PRODUCT_NAME
-----
QUANTITY UNIT_PRICE TOTAL_AMOUNT ORDER_STATUS
-----
2
Arjun Kumar
402 24-SEP-26

CUSTOMER_ID
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT
-----
PRODUCT_NAME
-----
QUANTITY UNIT_PRICE TOTAL_AMOUNT ORDER_STATUS
-----
Smartphone
1      25000      25000 Placed
```

9. Customer-wise Order Summary

```

SQL> SELECT
2   c.Customer_ID
3   c.First_Name || ' ' || c.Last_Name AS Customer_Name,
4   COUNT(o.Order_ID) AS Total_Orders,
5   SUM(o.Total_Amount) AS Total_Purchase_Amount
6 FROM Customer c
7 JOIN Orders o
8   ON c.Customer_ID = o.Customer_ID
9 GROUP BY
10  c.Customer_ID,
11  c.First_Name,
12  c.Last_Name
13 ORDER BY c.Customer_ID;

```

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_PURCHASE_AMOUNT

1		
Renu Srinivasan	1	55000
2		
Arjun Kumar	1	25000

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_PURCHASE_AMOUNT

3		
Priya Kumar	1	1598
4		
Rahul Sharma		

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_PURCHASE_AMOUNT

1		2500
5		
Keerthana Mohan	1	650